

Project Report: Guild Voting System

Group Members

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1. System Overview

The Guild Voting System is a web-based platform developed to streamline and secure elections within educational institutions. The system differentiates between student and administrator roles:

- **Students:** Can securely vote in ongoing elections and view tally results.
- **Administrators:** Add or manage students, block or enable voting for students, promote students to admins, and oversee election activities.
- Both user roles can personalize their experience with light/dark theme options.

2. Database Structure

Below is an outline of the primary database tables used in the system, showcasing their columns and relationships.

2.1 eligible_students

Column	Type	Description
id	INT, PK, AUTO_INC	Unique record ID
student_number	VARCHAR(20), UQ	Student number
full_name	VARCHAR(100)	Student's legal/full name
official_email	VARCHAR(150), UQ	Official student email
is_claimed	TINYINT(1)	If account is claimed
added_at	DATETIME	Time added

2.2 users

Column	Type	Description
id	INT, PK, AUTO_INC	User's unique ID
student_number	VARCHAR(20), UQ	Student number
full_name	VARCHAR(100)	User's full name
email	VARCHAR(150), UQ	Email address
password_hash	VARCHAR(255)	Hashed password
role	ENUM('student','admin')	User role
is_active	TINYINT(1)	Active status
registered_at	DATETIME	Registration timestamp

2.3 elections

Column	Type	Description
id	INT, PK, AUTO_INC	Election ID
title	VARCHAR(200)	Title of election
description	TEXT	Details
status	ENUM('pending','open','closed')	State of election
created_by	INT (FK: users.id)	Creator user ID
created_at	DATETIME	Creation time

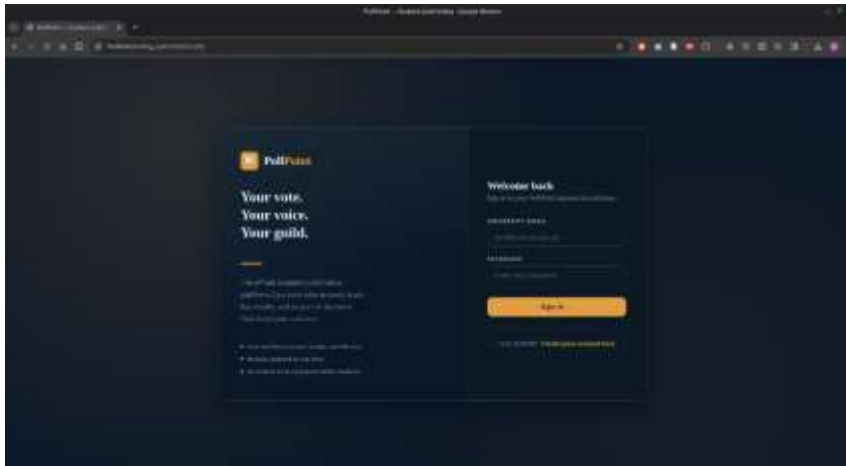
2.4 votes

Column	Type	Description
id	INT, PK, AUTO_INC	Vote record ID
student_id	INT, FK: users.id	Student who voted
election_id	INT, FK: elections.id	Election voted in
candidate_id	INT, FK: candidates.id	Candidate voted for
voted_at	DATETIME	Timestamp of vote

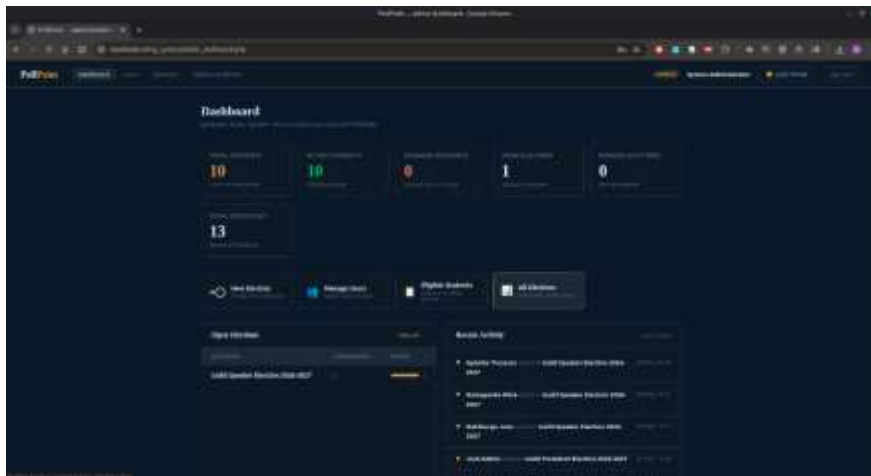
Note: Foreign keys enforce relational integrity and allow for secure and accurate tracking of votes and election management.

3. Sample Screenshots

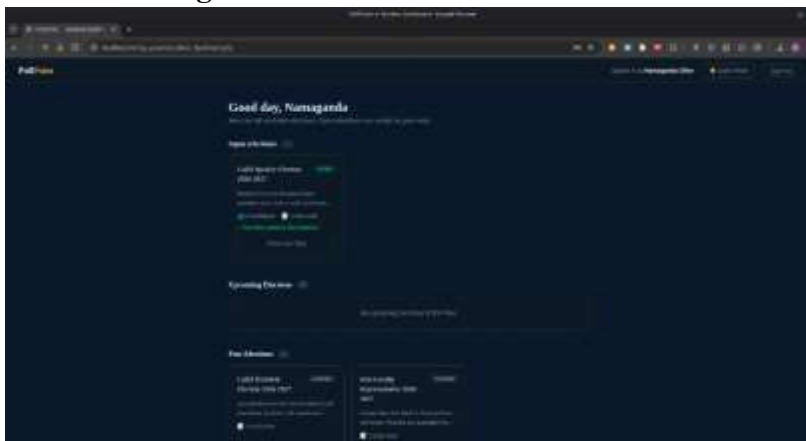
1. Login Screen



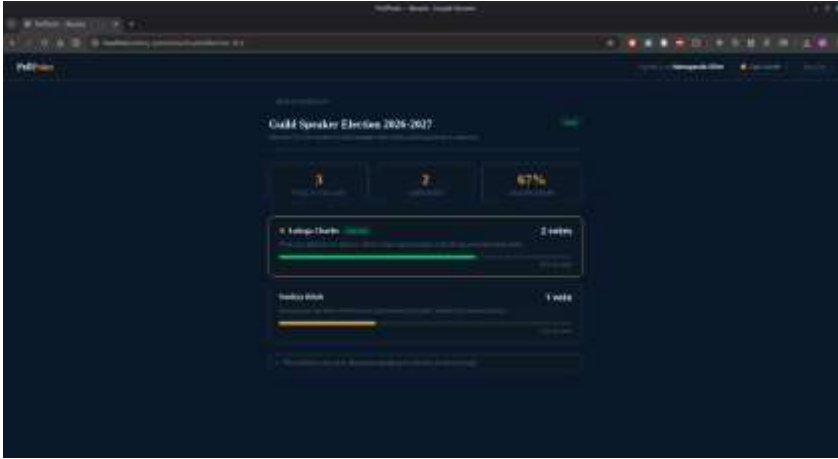
2. Admin Dashboard



3. Student Voting Interface



4. Vote Tally/Results Page



5. User Management (Admin)

The screenshot shows the PyCharm IDE interface. The top toolbar includes buttons for Run, Debug, and Test. The 'Run' button is highlighted. Below the toolbar, the 'Run' tool window is open, displaying a table of test results. The table has columns for Test Name, Status, Duration, and other details. The tests listed are 'add_numbers', 'multiply_numbers', 'divide_numbers', 'power_of_two', 'prime_number', 'palindrome_string', 'anagram_string', and 'is_leap_year'. All tests are marked as 'Passed' with a green status icon. Below the table, there is a section for 'Test Results Summary' showing 'Passed: 8/8' and 'Failed: 0/0'.

Test Name	Status	Duration	Other Details
add_numbers	Passed	0.001s	100% Passed
multiply_numbers	Passed	0.001s	100% Passed
divide_numbers	Passed	0.001s	100% Passed
power_of_two	Passed	0.001s	100% Passed
prime_number	Passed	0.001s	100% Passed
palindrome_string	Passed	0.001s	100% Passed
anagram_string	Passed	0.001s	100% Passed
is_leap_year	Passed	0.001s	100% Passed

Test Results Summary

Passed: 8/8
Failed: 0/0

4. Challenges & Lessons Learned

Challenges:

- Spent significant time designing an appropriate and scalable database schema to handle users, votes, and roles efficiently.
- Juggled project work alongside exam preparation, making time management critical.
- Ensured cross-platform compatibility by testing code on both Linux and Windows environments where we had some small server/configuration differences.

Lessons Learned:

- The importance of planning out database relationships early.
- Regular testing on different environments helps catch subtle issues.
- Clear role separation (student/admin) enhances both security and usability.